

# 一个三元不定方程

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问题：求解不定方程

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} = 4$$

的最小正整数解.

解：通分原方程得到

$$a^3 - 3a^2b - 3a^2c - 3ab^2 - 5abc - 3ac^2 + b^3 - 3b^2c - 3bc^2 + c^3 = 0$$

置变换

$$a = \frac{56-x+y}{56-14x}, b = \frac{56-x-y}{56-14x}, c = \frac{6x+28}{7x-28}$$

得到椭圆曲线

$$E: y^2 = x^3 + 109x^2 + 224x$$

其逆变换为：

$$x = -\frac{28(a+b+2c)}{6a+6b-c}, y = \frac{364(a-b)}{6a+6b-c}$$

$E$  有一有理点  $P(-100, 260)$ , 对其做椭圆曲线上的加法运算, 可得到一系列有理解, 代入  $(a, b, c)$  可得原方程可能的解. 因为我们需要的是最小正整数解, 因此需要对解“约化”, 即乘以三数分母的最小公倍数以去分母, 如果解有一个为负, 就需要继续加, 直到结果符合条件.

以下为计算结果, 一直到  $9P$  我们才找到了最小解.

$$1P = (-100, 260) \rightarrow (a, b, c) = \left(\frac{2}{7}, -\frac{1}{14}, \frac{11}{14}\right) \xrightarrow{\text{约化}} (4, -1, 11)$$

$$2P = \left(\frac{8836}{25}, -\frac{950716}{125}\right) \rightarrow (a, b, c) = \left(\frac{1357}{840}, -\frac{366}{245}, \frac{1033}{1176}\right) \xrightarrow{\text{约化}} (9499, -8784, 5165)$$

$$3P = \left(-\frac{731025}{11881}, \frac{527529870}{1295029}\right) \rightarrow (a, b, c) = \left(\frac{975227}{1704542}, -\frac{336013}{1063622}, \frac{4053482}{5449843}\right) \xrightarrow{\text{约化}} (679733219, -375326521, 883659076)$$

$$\begin{aligned}
4P &= \left( \frac{561561391876}{8356702225}, -\frac{687837762272090924}{763927933898375} \right) \rightarrow \\
(a, b, c) &= \left( \frac{64375537326}{62482243915}, -\frac{66892284997}{66560358480}, \frac{69295307953}{71095040016} \right) \\
&\xrightarrow{\text{约化}} (6696085890501216, -6531563383962071, 6334630576523495) \\
5P &= \left( -\frac{425869857827702500}{15192076294211881}, \frac{448412887098983162164732300}{1872516697802137088411221} \right) \rightarrow \\
(a, b, c) &= \left( \frac{132826132539336599}{184128998198450774}, -\frac{13113991869578878}{37835009114259847}, \frac{40958480975543891}{65508983481381734} \right) \\
&\xrightarrow{\text{约化}} (5824662475191962424632819, -2798662276711559924688956, 5048384306267455380784631) \\
6P &= \left( \frac{252785840525963937198721}{13225347684085115955600}, -\frac{343764653760831645784970282294394569}{1520934975898868459000385442296000} \right) \rightarrow \\
(a, b, c) &= \left( \frac{1262869893606655032359641}{1412814438402889060433160}, -\frac{455681313378081801431401}{366738964125570802591560}, \right. \\
&\quad \left. \frac{1887024778310166869949126}{1399191148527364313634247} \right) \xrightarrow{\text{约化}} (287663048897224554337446918344405429, \\
&\quad -399866258624438737232493646244383709, 434021404091091140782000234591618320) \\
7P &= \left( -\frac{1872773018543093075805479817148900}{163350615997049698719631653803161}, \right. \\
&\quad \left. \frac{211390151297491981533823798000243842339853402161620}{2087762847145230771938050768331341412318712353341} \right) \rightarrow \\
(a, b, c) &= \left( \frac{485999570706736171999361687151362}{623660861246956197068644226999047}, -\frac{97299499707397042459551895751641}{623488630867002224845371906564614}, \right. \\
&\quad \left. \frac{128131170448868594051599857623171}{340062084186904674899770096664054} \right) \xrightarrow{\text{约化}} (3386928246329327259763849184510185031406211324804, \\
&\quad -678266970930133923578916161648350398206354101381, \\
&\quad 1637627722378544613543242758851617912968156867151) \\
8P &= \left( \frac{8304830821343520148948729081523501917420036}{1462178082526064533227321540886890051021025}, \right. \\
&\quad \left. -\frac{124668752376211382766812793520103329799089543359986485854396841484}{1768073864797815348625142542347514156118825216532230801892732625} \right) \rightarrow \\
(a, b, c) &= \left( \frac{9725799744957614839011216987101754625826197}{11327874209838418438467575524552717247648160}, \right. \\
&\quad \left. -\frac{1534957039510080155063626001967221164131246}{298740362138731751304699903310656685729205}, \frac{1745576369976748612001103416037960248713633}{330631335359131425236078854342915230641376} \right) \\
&\xrightarrow{\text{约化}} (343258303254635343211175484588572430575289938927656972201563791, \\
&\quad -2054217703980198940765993621567260834791816664149006217306067776, \\
&\quad 2110760649231325855047088974560468667532616164397520142622104465)
\end{aligned}$$

$$\begin{aligned}
9P = & \left( - \frac{66202368404229585264842409883878874707453676645038225}{13514400292716288512070907945002943352692578000406921}, \right. \\
& \left. \frac{58800835157308083307376751727347181330085672850296730351871748713307988700611210}{1571068668597978434556364707291896268838086945430031322196754390420280407346469} \right) \rightarrow \\
(a, b, c) = & \left( \frac{652194680638776317370751188686261401138670498641722947}{826345176768069653846031682295795260307016241032351542}, \right. \\
& \frac{72627067629030455550043880234643101653454184810448427}{385489402115598358968822193146517732601759618776822382}, \\
& \left. \frac{18811002229321433251069036843190834369329875858835562}{841819787025663175191882291647234536827567920526661363} \right) \xrightarrow{\text{约化}} \\
& (154476802108746166441951315019919837485664325669565431700026634898253202035277999, \\
& 36875131794129999827197811565225474825492979968971970996283137471637224634055579, \\
& 4373612677928697257861252602371390152816537558161613618621437993378423467772036)
\end{aligned}$$